

Traffic Congestion and Urban Heat: Evidence from Freeway Incidents in California

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Abstract

Traffic congestion is a plausible but unquantified contributor to localized urban heat, a costly externality for the many who live and work near major roads. Yet its short-run causal effect on local surface temperature has not been credibly identified: existing estimates are associational or rely on broad natural experiments. We estimate it for California freeways using unplanned accidents as an instrument for congestion. The central threat is reverse causality: because congestion raises accident risk, accidents are in general endogenous to the congestion they are meant to instrument—a bias we document with an event study and remove by restricting the instrument to accidents that strike while the segment is free-flowing, before congestion has formed. We measure land surface temperature from NASA ECOSTRESS at 70-meter resolution and exploit its varying overpass time to match each reading to same-hour congestion. Our central estimate is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.08°C over the following two hours, against flat pre-trends. Scaled by the accident’s effect on congestion, this implies roughly 0.5 to 0.8°C per 10 mph of speed reduction, somewhat above the prior associational estimate. The effect is statistically significant in the near-road band and declines significantly with distance from the road, tracing how far a localized transport externality reaches into the surrounding residential space.

Keywords: urban heat island, traffic congestion, land surface temperature, instrumental variables, freeway incidents, ECOSTRESS, California

JEL Classification: Q54, R41, C26

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1 Introduction

How far does the heat generated by traffic reach into the surrounding city? Extreme heat is the leading cause of weather-related deaths in the United States, exceeding floods, tornadoes, and hurricanes in average annual fatalities (National Weather Service, 2024). Urban heat, the systematic warmth of developed areas relative to their surroundings, carries well-documented health and energy costs (Heaviside et al., 2017; Santamouris et al., 2015), and its *structural* drivers, the pavement, rooftops, and sparse vegetation that set a neighborhood’s baseline temperature, are extensively studied (Oke, 1973; Gago et al., 2013). Much less is known about its *time-varying* drivers, and in particular about traffic congestion, which fluctuates hour to hour over a fixed road network that millions of people live beside: about 11 million Californians reside within one kilometer of the freeway segments we study, 4.7 million of them within 500 meters. Whether congestion measurably heats these neighborhoods, and how far that heat reaches, is an understudied spatial question.

The mechanism is well grounded: vehicles in stop-and-go operation burn more fuel and release more waste heat per mile than vehicles at cruise speed (Barth and Boriboonsinsin, 2008), and the anthropogenic heat-flux literature identifies vehicular traffic as a substantial, commute-peaked component of the urban energy balance (Sailor and Lu, 2004; Sailor, 2011; Quah and Roth, 2012; Ichinose et al., 1999). But causal identification of the congestion-to-heat link is limited. The closest evidence (Lee and Berkelhammer, 2025) correlates bus speeds with satellite temperature in Chicago and finds a 0.36°C rise per 10 mph of slowdown, an estimate that is associational; quasi-experimental alternatives rely on shocks, such as highway expansions (Yang et al., 2026) and COVID-19 mobility restrictions (Tran et al., 2026), that move land use, economic activity, and human presence alongside traffic.

We estimate the effect of freeway congestion on nearby surface temperature using unplanned freeway accidents logged in the Caltrans Performance Measurement System (PeMS) as an instrument for congestion. The central threat to this design is *reverse causality*: congestion itself raises accident risk, so we restrict the instrument to *free-flow-onset* accidents,

those that strike while the segment is free-flowing and therefore cannot have been caused by congestion. An event study validates the restriction: the speed performance index is flat before a free-flow-onset accident and drops sharply when the accident occurs, and surface-temperature pre-trends are flat (Section 4.2). We measure the temperature response with NASA ECOSTRESS, whose 70-meter resolution and varying overpass times let each land surface temperature (LST) reading be matched to a PeMS congestion reading from the same hour, and we estimate the response in concentric distance bands around the freeway, separating the temperature of the road itself from the temperature experienced by nearby households, in the spirit of Currie and Walker (2011).

We report the estimates as a progression rather than a single headline number. Within detector-by-date-by-hour variation, the OLS association is small and wrong-signed, so the raw correlation cannot be read as the effect of congestion on temperature. The naive instrument using all accidents yields a precise, correctly signed estimate that inherits this contamination. Our central estimate is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.08°C over the following two hours, against flat pre-trends. Expressed per unit of congestion through two-stage least squares, this implies roughly 0.5 to 0.8°C per 10 mph reduction in speed, somewhat above the associational estimate in Lee and Berkelhammer (2025). The effect is correctly signed at every distance band, is statistically significant near the road, and declines significantly with distance from it.

These magnitudes have economic content beyond their size. The delay and pollution costs of congestion are widely studied and, in places, priced; the thermal channel is an additional local cost that no current instrument prices, borne by those who live beside the network and concentrated at commute hours, when congestion and outdoor exposure coincide. It is also a cost that existing policy already moves: because the estimates are identified from accident-induced congestion episodes, they measure what rapid incident detection and clearance (programs transportation agencies operate primarily to recover delay) also buy back in neighborhood heat.

The paper makes three contributions. First, it provides a credibly identified estimate of the short-run effect of traffic congestion on local surface temperature, where prior evidence is associational. Second, it shows that accidents, used as instruments for congestion precisely because they appear exogenous (Beland and Brent, 2018), are systematically endogenous to the congestion they are meant to instrument, and that conditioning on the pre-accident state removes the bias. Third, it maps how far the congestion-heat externality reaches from the road into the surrounding residential fabric, separating the on-road effect from the temperature experienced by nearby households.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature; Section 3 describes the data; Section 4 presents the research design, including the event study; Section 5 reports the results; Section 6 reports the robustness checks central to identification, with further checks in Appendix; and Section 7 concludes.

2 Related Literature

This paper connects several bodies of work: the urban heat island literature, a small literature on traffic congestion and local temperature, and the causal economics literature on traffic externalities.

Urban heat islands arise when vegetated, permeable surfaces are replaced with impervious materials that absorb and retain solar radiation while eliminating evaporative cooling (Oke, 1973; Gago et al., 2013; Declet-Barreto et al., 2013). The stakes are well quantified: Heaviside et al. (2016) attribute roughly half of heat-related deaths in the 2003 West Midlands heatwave to the urban heat island, Tuholske et al. (2021) document a near-doubling of urban population exposure to extreme heat since the 1980s, and Santamouris et al. (2015) estimate that urban overheating raises building cooling loads by 13 percent on average. This literature centers on structural features of the built environment that vary slowly over time, measured at scale through thermal infrared remote sensing (Li et al., 2020); we study

a time-varying driver, congestion, using the 70-meter, varying-overpass-time ECOSTRESS instrument (Fisher et al., 2020) to match temperature to traffic conditions hour by hour.

Direct evidence linking congestion to local temperature is scarce. Husni et al. (2022) document a 7°C temperature increase associated with traffic queue rates using IoT sensors in Indonesia; Yang et al. (2026) find that Bay Area highway expansions exacerbate urban–rural temperature differentials in MODIS data; and Lee and Berkelhammer (2025) estimate a 0.36°C increase in surface urban heat island intensity per 10 mph decline in Chicago bus speeds. These designs are associational; our contribution is causal identification of the same relationship, at finer spatial resolution and statewide scale.

A larger economics literature establishes that traffic imposes substantial external costs on infant health, productivity, and housing values (Greenstone and Gayer, 2009; Knittel et al., 2016; Isaksen and Johansen, 2021; Graff Zivin and Neidell, 2012; Chay and Greenstone, 2005). Chay and Greenstone (2003) establish the foundational causal link between vehicle-related pollution and infant health; Xia et al. (2023) study the within-city effects of congestion on NO_2 and CO , the closest pollution-side analog to our congestion-temperature design; and Currie and Walker (2011) compare birth outcomes for mothers living within two kilometers of a toll plaza with those two to ten kilometers away, before and after the introduction of E-ZPass reduced idling and congestion at the plazas, finding improved outcomes for the nearby mothers. We extend this literature to the temperature channel, which operates through the same waste heat and exhaust as the pollution channel but along a thermal rather than chemical pathway. To instrument for congestion itself, rather than infrastructure (where Duranton and Turner (2011) established the canonical design), prior work has used transit strikes (Anderson, 2014) and other transitory shocks to traffic; unplanned freeway accidents offer finer spatial and temporal resolution than these, located to the detector segment and timestamped to the minute. This approach draws on a broad tradition in economics that exploits the idiosyncratic short-run timing of natural shocks, weather most prominently, as a source of exogenous variation (Miguel et al., 2004; Deschênes and Greenstone, 2007).

Two further literatures bear directly on the design. Accidents have been used in economics as a source of exogenous variation in traffic: studying the effect of congestion on crime, Beland and Brent (2018) report a specification that instruments extreme traffic in Los Angeles with Caltrans-logged accidents delaying traffic by more than an hour, drawn from the same PeMS detector data we use. Transportation research, however, shows that freeway crash risk rises with congestion and can be predicted from detector readings in the minutes before a crash (Abdel-Aty et al., 2004; Yeo et al., 2013), so accidents cannot be assumed exogenous to the congestion around them. That the instrument appears in published work on our own data source is what makes the endogeneity we document consequential beyond this setting. And a body of work identifies near-road exposure from wind direction relative to the source (Anderson, 2020; Herrnstadt et al., 2021); our upwind-downwind comparison follows that logic, applied to heat rather than to pollution.

Finally, the paper’s spatial object, the distance gradient of the temperature response, connects to a regional-science tradition tracing how localized disamenities decay with distance from their source: Currie et al. (2015) measure the reach of toxic-plant emissions into surrounding housing markets, Seo et al. (2014) estimate distance-band hedonic gradients of highway disamenity, and Brinkman and Lin (2024) value the local quality-of-life costs of urban highways. We measure the same kind of spatial decay for a transport externality in physical rather than price terms, asking how far congestion-generated heat extends into the residential fabric around the network.

3 Data

3.1 Traffic Congestion and Incident Logs: PeMS

Traffic data come from the Caltrans Performance Measurement System (PeMS; California Department of Transportation, 2026), which operates a network of traffic detectors embedded in California freeway lanes. Each detector records flow (vehicles per hour), occupancy (share

of time a vehicle is present over the sensor), and speed (miles per hour) at five-minute intervals. Detectors are spaced approximately every half mile along major California freeways and are organized into stations that aggregate across lanes. We obtain station-level hourly data and station metadata (GPS coordinates, freeway route, and direction of travel) for eight Caltrans districts from June 2018 through March 2026.¹ Figure 1 maps the detectors in the analysis sample across these eight districts.

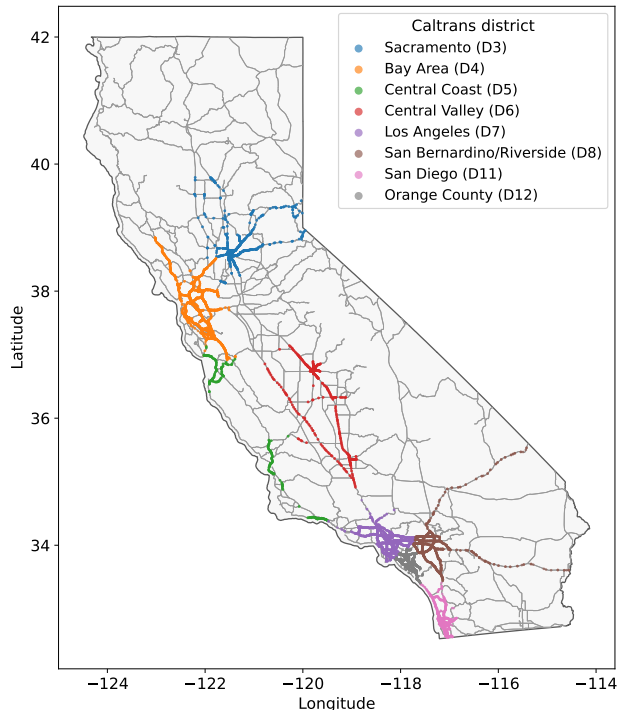


Figure 1: PeMS freeway detectors in the analysis sample, by Caltrans district. The sample spans the eight districts covering California’s major metropolitan areas; detectors trace the instrumented freeway network in each metro.

The primary congestion measure is the *speed performance index*, a standard speed-based congestion metric (He et al., 2016) equal to the ratio of realized speed to free-flow

¹We include the eight Caltrans districts covering California’s major metropolitan areas: District 3 (Sacramento), District 4 (Bay Area), District 5 (San Luis Obispo/Central Coast), District 6 (Fresno/Central Valley), District 7 (Los Angeles), District 8 (San Bernardino/Riverside), District 11 (San Diego), and District 12 (Orange County). Districts 1 (North Coast), 2 (Northern California), 9 (Eastern Sierra), and 10 (Central Sierra) are excluded because they cover predominantly rural and mountainous terrain with sparse freeway detector networks and limited urban heat island exposure.

speed:

$$\text{SPI}_{it} = \frac{\text{Speed}_{it}}{\text{Free-Flow Speed}_i}$$

where i indexes the PeMS detector station and t indexes the hour of the ECOSTRESS overpass. A value of 1 indicates free-flow conditions and values approaching 0 indicate near-standstill conditions, so the index *falls* as congestion worsens. Free-flow speed $\bar{v}_i$ is defined empirically as the 85th percentile of observed speeds at station i during overnight hours (10pm–5am) and weekends, when congestion is minimal. This percentile-based definition makes the index comparable across segments with different posted speed limits and traffic compositions.

For the incident instrument, we use PeMS incident logs, which record the timestamp, segment, type, and duration of each incident logged on California freeways. PeMS classifies incidents into a range of categories, the most common of which are traffic collisions, traffic hazards, animal hazards, vehicle fires, road and weather conditions, and assists to Caltrans operations. We restrict the instrument to traffic accidents, which are the most plausibly exogenous to neighborhood characteristics and environmental conditions. Weather-related incidents are excluded because precipitation simultaneously raises accident rates and suppresses LST, creating a mechanical negative correlation that would bias the instrument. Traffic hazards, animal hazards, vehicle fires and debris, and planned events such as construction assists and road closures are excluded because they either do not reliably generate congestion or are not unplanned. We discuss these restrictions further in Section 4.3.

The PeMS incident archive is largely complete over the study period, except for March-early July 2025, where the feed was not reliably archived.² We treat the instrument as undefined on dates lacking reliable incident records (missing, empty, or anomalously sparse) and *drop* ECOSTRESS overpasses falling in those windows. The 2025 gap therefore removes roughly four months of overpasses from the estimation sample. Because the gap is a data-

²The remaining gaps are minor and isolated: a single day in June 2019, a cluster of days in January 2020, and scattered individual dates, together accounting for under two percent of days in the study period. Incident coverage in all other months is complete.

archiving artifact unrelated to traffic, weather, or neighborhood conditions, this attrition is plausibly unrelated to the outcome and is not expected to bias the estimates; we report the affected sample counts in Table 1.

3.2 Land Surface Temperature: ECOSTRESS

Land surface temperature data come from the NASA ECOSTRESS ECO_L2T_LSTE Version 2 product, distributed by the NASA Land Processes Distributed Active Archive Center and accessed via AppEEARS (Hulley and Hook, 2022). ECOSTRESS is a thermal infrared instrument on the International Space Station (ISS) that has provided observations since July 2018 at 70-meter spatial resolution (Fisher et al., 2020). The 70-meter resolution is substantially finer than the 1-kilometer MODIS data used in prior UHI research (Chakraborty et al., 2019) and allows LST measurement at the scale of individual freeway segments and adjacent land parcels.

The defining temporal property of ECOSTRESS for this study is its ISS orbit. Unlike sun-synchronous satellites that overpass at a fixed local solar time (Landsat 8 overpasses at approximately 10am; MODIS at fixed morning and afternoon times), the ISS precesses through a range of local overpass times at any given California location. This means successive ECOSTRESS overpasses of the same segment occur at different times of day, generating exogenous variation in the hour of LST observation; Figure 2 plots the distribution of overpass times across the 24-hour day. We exploit this variation to match each LST observation to a contemporaneous PeMS congestion reading at the overpass hour. The hour-of-day variation in overpass time also underlies the event study evidence described in Section 4.2.

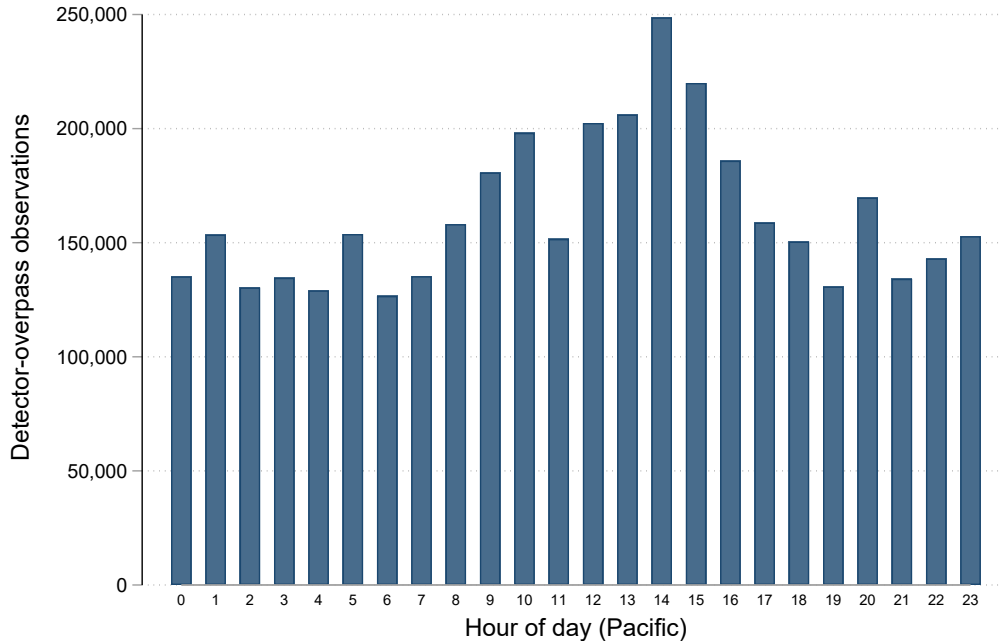


Figure 2: Distribution of ECOSTRESS overpass times by hour of day (Pacific) in the matched sample. Because the International Space Station orbit is not sun-synchronous, overpasses occur across all 24 hours rather than at a fixed local solar time. Identification comes from accident timing, with hour-of-day fixed effects absorbing the solar cycle; the varying overpass time matters because it yields observations at the hours when accidents occur.

We download four layers from ECO_L2T_LSTE: LST (in Kelvin), LST_err (retrieval uncertainty), QC (quality control flags), and cloud (cloud mask). We apply quality filters to exclude pixels where the LST retrieval is unreliable. Three sources of retrieval failure motivate these filters. First, clouds absorb and re-emit thermal radiation before it reaches the sensor, so a cloudy pixel records the temperature of the cloud rather than the ground surface; we exclude pixels flagged as cloudy by the cloud mask. Second, the retrieval algorithm corrects the raw thermal signal for atmospheric absorption and scattering, but this correction can fail under unusual water vapor or aerosol conditions (including, potentially, haze from heavily congested corridors), producing biased LST values; we exclude pixels with QC flag values indicating failed or degraded retrievals. Third, the algorithm converts radiance to temperature using assumed surface emissivity, and high uncertainty in that assumption produces inaccurate temperatures. We therefore drop pixels whose re-

trieved LST falls outside a physically plausible range (255–345 K, about -18 to 72°C), which removes implausible retrievals that survive the cloud and quality-control screens. The per-pixel retrieval-uncertainty layer (LST_err) is downloaded and available but is not applied as a screen in the baseline specification. Using corrupted pixels would add noise to the outcome variable and could introduce systematic bias if retrieval quality is correlated with congestion levels. After applying these filters, the outcome variable for observation (i, t) is the mean LST across all pixels passing quality filters within the 500-meter buffer of detector i at overpass t . The spatial footprint is restricted to a 2-kilometer buffer around the California state highway network, encompassing all PeMS detector locations and the immediately surrounding land.

Because ECOSTRESS is a relatively young instrument, coverage in the earliest months of our study period is incomplete. ECOSTRESS experienced two anomalies of its onboard mass storage unit (MSU) that suspended data acquisition: the primary unit failed on September 29, 2018 and acquisition resumed only after the instrument was switched to its backup unit on December 5, 2018, and a second anomaly halted acquisition again between March 14 and May 15, 2019 (NASA Land Processes Distributed Active Archive Center, 2024). Our panel therefore contains no LST observations for October through December 2018 and reduced coverage in the spring of 2019. Because these gaps arise from instrument hardware status rather than from any feature of traffic, weather, or location, the lost observations are unrelated to the outcome and affect only the early-period sample size, not the identifying variation.

3.3 Sample Construction

The unit of observation is a freeway detector station i crossed with an ECOSTRESS overpass t . A valid observation requires: (1) at least five ECOSTRESS pixels passing quality filters within the buffer of the detector; (2) a valid PeMS congestion reading within the same calendar hour as the overpass timestamp; and (3) that congestion reading passing the data-

quality filter, at least 50 percent of the hour’s five-minute intervals observed (`pct_observed` ≥ 50), to reduce measurement error from detector malfunctions.

3.3.1 Outcome: distance bands around the freeway

The outcome is land surface temperature in concentric distance bands around the detector, each the mean of quality-filtered ECOSTRESS pixels within a given radius at overpass t . We construct the 100-, 250-, 500-, and 1,000-meter buffers, and from them three disjoint annular rings (0–100, 100–500, and 500–1,000 meters) for the distance gradient. The 100-meter band captures near-road temperature, where traffic-related heat is most concentrated. The band spacing follows the near-road pollution literature, which finds concentrations decaying to background within 150 to 300 meters and detectable to roughly 500 meters (Health Effects Institute, 2010); we use those distances to set the geometry of the bands, not as evidence about how far heat itself travels, since a pollutant disperses as a near-conserved tracer while heat is shed to the air and to the sky far faster. The outer bands capture temperature in the surrounding residential area, the exposure of most direct welfare interest, in the spirit of Currie and Walker (2011), who study residential outcomes for mothers within two kilometers of toll plazas. At 70-meter resolution a 500-meter radius contains about 160 pixels, yielding a stable average; we take the 500-meter buffer as the descriptive baseline, while the preferred causal estimate uses the near-road 100-meter band. Section 5 reports every band, and the gradient across the disjoint rings maps how far the effect reaches.

Table 1 reports the observations lost at each of these filtering stages. The binding step is the same-hour match, which retains about 46 percent of quality-passing overpasses: a freeway detector reports a usable speed only intermittently (some never report speed at all, and a working detector is online at a given overpass hour only about half the time), so the matched panel is the intersection of two independent data streams rather than a curated subsample. The match rate is essentially flat across all twenty-four hours of the day, so this attrition costs statistical power without selecting on the overpass-time variation the design

exploits; and because detector reporting reliability and cloud cover are plausibly unrelated to the idiosyncratic timing of freeway accidents, we do not expect it to threaten identification.

Table 2 reports summary statistics for the full analysis sample and, separately, for the free-flow-onset-treated overpasses that identify the clean estimate. Treatment is rare by design: accidents are infrequent, so only 5,378 of the 3.9 million detector-overpasses carry a free-flow-onset accident in the prior two hours. The remaining observations are not incidental—they are the control pool, and identification comes from comparing treated overpasses to control overpasses of the same detector, date, and hour-of-day. Two patterns are worth noting. The typical overpass is close to free-flow (mean speed performance index 0.92), so the congestion variation that identifies the effect lies in the lower tail of the distribution rather than at the mean; and the treated overpasses are modestly more congested than the full sample (index 0.90 versus 0.92, with higher traffic flow), the descriptive counterpart of the first stage. Surface temperature averages 22.8°C with wide dispersion (standard deviation 11.5°C), reflecting the all-hours, multi-year, statewide ECOSTRESS coverage that the detector, date, and hour-of-day fixed effects absorb.

Table 1: Sample construction

	Observations	Detectors	% of row 1
ECOSTRESS overpass observations	9,045,115	11,180	—
with ≥ 5 quality-passing pixels (500 m buffer)	8,910,223	11,178	98.5
matched to a same-hour PeMS congestion reading	4,124,564	9,323	45.6
excluding low-coverage incident days	3,890,544	9,287	43.0
Estimation panel	3,890,544	9,287	43.0
with non-missing weather controls	3,702,727	8,889	40.9

Notes: The unit of observation is a detector $\times$ ECOSTRESS overpass. Row 1 counts overpasses with a valid LST extraction (June 2018–March 2026; after pixel-level cloud, quality-control, and 255–345 K masking). The large drop at the same-hour PeMS match (row 3) reflects detector intermittency, not selection: the match rate is flat across all 24 hours of the day, so the attrition is orthogonal to the overpass-time variation the design exploits. Row 4 drops overpasses on dates with missing or anomalously sparse CHP incident records (the March–July 2025 feed gap and scattered days), which cannot be coded incident = 0. The final row is the sample for the preferred weather-controlled specification.

Table 2: Summary statistics

	Full sample		Free-flow-onset treated	
	Mean	SD	Mean	SD
Land surface temp (°C)	22.76	11.46	22.46	10.95
Speed performance index (speed / free-flow)	0.92	0.12	0.90	0.15
Speed (mph)	63.49	8.72	62.29	10.73
Free-flow speed (mph)	68.65	2.23	68.96	1.48
Traffic flow (veh/hr)	2,674	2,054	3,020	1,924
ECOSTRESS pixels in buffer	142.5	35.5	143.0	34.9
Observations	3,890,544		5,378	

Notes: Unit of observation is a PeMS detector $\times$ ECOSTRESS overpass, June 2018–March 2026. “Full sample” is the estimation panel; “free-flow-onset treated” is the subsample of overpasses with a free-flow-onset accident (one striking the segment while it was free-flowing, speed performance index ≥ 0.90) in the prior two hours, the variation that identifies the clean estimate. The speed performance index is realized speed divided by free-flow speed (1 = free-flow; lower = more congested).

4 Research Design

4.1 Identification Problem

The parameter of interest is the causal effect of traffic congestion on land surface temperature. Let LST_{it} denote surface temperature at detector location i at overpass time t , and let SPI_{it} denote the speed performance index on the corresponding segment. The population relationship of interest is:

$$LST_{it} = \alpha_i + \mu_d + \lambda_h + \delta \cdot SPI_{it} + \mathbf{X}'_{it}\gamma + u_{it}$$

where α_i are detector fixed effects, μ_d are date fixed effects (where d indexes the calendar date of overpass t), λ_h are hour-of-day fixed effects (where h indexes the Pacific-time hour of the overpass, $h \in \{0, 1, \dots, 23\}$), $\mathbf{X}_{it}$ is a vector of local weather controls, and u_{it} is an error term. The coefficient δ is the causal effect of a one-unit decrease in the speed performance index (corresponding to a shift from free-flow to fully congested conditions) on surface temperature in degrees Celsius. We measure LST_{it} in concentric distance bands

around the segment (Section 3.3.1); the near-road 100-meter band, where the mechanism places the response, is the preferred outcome and the 500-meter buffer a descriptive baseline; Section 5 reports δ for each band, from the roadway out to one kilometer.

Identification is within-segment, within-date, and within-hour: α_i absorbs time-invariant location characteristics (pavement, land cover, urban density), μ_d absorbs common daily shocks such as heat waves without eliminating within-day variation in overpass time, and λ_h absorbs the solar cycle. The controls $\mathbf{X}_{it}$, air temperature, precipitation, relative humidity, and wind speed at the detector and overpass hour, drawn from the ERA5-Land hourly reanalysis (Muñoz Sabater et al., 2021; Muñoz Sabater, 2019), absorb local weather not captured by the date fixed effects.

Even within these cells, SPI_{it} is endogenous: clear skies raise both solar load and traffic volumes, biasing OLS upward, while heat-induced travel avoidance works in the opposite direction, so the sign of the net bias is ambiguous *ex ante*. This motivates the IV approach. The instrument we use, freeway accidents, raises a distinct identification concern of its own, reverse causality, since congestion increases accident risk. We confront it directly with the event study that follows.

4.2 Event Study

Because an accident-based instrument is valid only if accident timing is exogenous to congestion, we begin the design with an event study that tests this directly and motivates the form of the instrument. It traces congestion and surface temperature in two-hour bins around an accident and asks two questions: is congestion already building before impact (reverse causality), and are there pre-accident trends in surface temperature that would violate the exclusion restriction? The answers lead us to restrict the instrument to free-flow-onset accidents.

4.2.1 Design

For each ECOSTRESS overpass we measure the elapsed time to each qualifying accident on the same detector within a ± 6 -hour window and assign the overpass to a two-hour lag bin $[\ell, u)$:

$$d_{\ell,u,it} = \mathbf{1}[t - \text{accident}_{it} \in [\ell, u)].$$

Negative lags are overpasses *before* an accident; the baseline group has no accident on the segment within ± 6 hours. We estimate

$$Y_{it} = \alpha_i + \mu_d + \lambda_h + \sum_{[\ell,u)} \phi_{\ell,u} d_{\ell,u,it} + \eta_{it} \quad (1)$$

separately for $Y_{it} = \text{SPI}_{it}$ (speed performance index) and $Y_{it} = \text{LST}_{it}$ (degrees Celsius), with detector, date, and hour-of-day fixed effects and standard errors clustered at the detector; we omit the weather controls $\mathbf{X}_{it}$ here, as the event study is a reduced-form diagnostic. The lead coefficients $\phi_{\ell,u}$ at negative lags are the diagnostic of interest, in speed-performance-index units (the index averages about 0.92).

4.2.2 All accidents: congestion rises before impact

Estimated on all accidents, the instrument is contaminated. Congestion is not flat before impact: the lead coefficients rise steadily over the hours before an accident (-0.007 , -0.023 , -0.060) and peak at the accident bin (-0.076), roughly symmetric around impact—the signature of reverse causality, with accidents occurring disproportionately during transient congestion (Appendix Figure 6). Surface-temperature leads, by contrast, are flat and insignificant, with the only significant coefficient at the accident bin ($+0.09^\circ\text{C}$; Appendix Figure 7): the exclusion restriction looks credible, but the all-accident instrument is not exogenous to congestion itself.

4.2.3 Free-flow-onset accidents: isolating exogenous variation

To separate exogenous from congestion-induced accidents, we classify each accident by the speed performance index on its segment in the hour before impact. A *free-flow-onset* accident strikes while the segment is free-flowing (pre-accident $\text{SPI} \geq 0.90$) and so cannot have been caused by congestion; a *congested-onset* accident strikes while the segment is already congested. Among accidents we can classify, the two groups are roughly equal in number. We re-estimate equation (1) with separate bin sets for the two groups in a single regression, so that congested-onset overpasses do not enter the free-flow-onset baseline (Figures 3–4).

The two groups behave very differently in congestion (Figure 3). Before a congested-onset accident, congestion ramps steeply ($-0.024, -0.069, -0.151$ across the three lead bins): the reverse causality is concentrated here. Before a free-flow-onset accident, congestion does the opposite: the segment is, if anything, increasingly free-flowing ($+0.009, +0.022, +0.029$) and then drops sharply at impact ($-0.014, t \approx -7$). Free-flow-onset accidents thus deliver a sharp, discontinuous congestion shock with no congestion building beforehand. The upward drift in those leads is not a congestion pre-trend but the mirror image of one: it reflects the persistence of the free-flowing state on which the instrument conditions, since the conditioning hour itself falls in the final lead bin. We therefore do not read the leads as validating the magnitude of the first stage, and we quantify the persistence directly with a pseudo-event placebo that reproduces the conditioning without an accident (Section 6). What the leads do establish is that congestion is not rising into these accidents, which is the reverse-causality threat the restriction exists to remove. Surface-temperature leads are flat for both groups (Figure 4); for the free-flow-onset group the three lead coefficients are individually and jointly indistinguishable from zero ($F(3, 9,250) = 0.25, p = 0.86$), consistent with the exclusion restriction. The free-flow-onset accident bin shows a positive temperature response at impact, quantified in Section 5.

These patterns motivate the rest of the design: we use free-flow-onset accidents as the instrument, because their congestion variation is not contaminated by reverse causality.

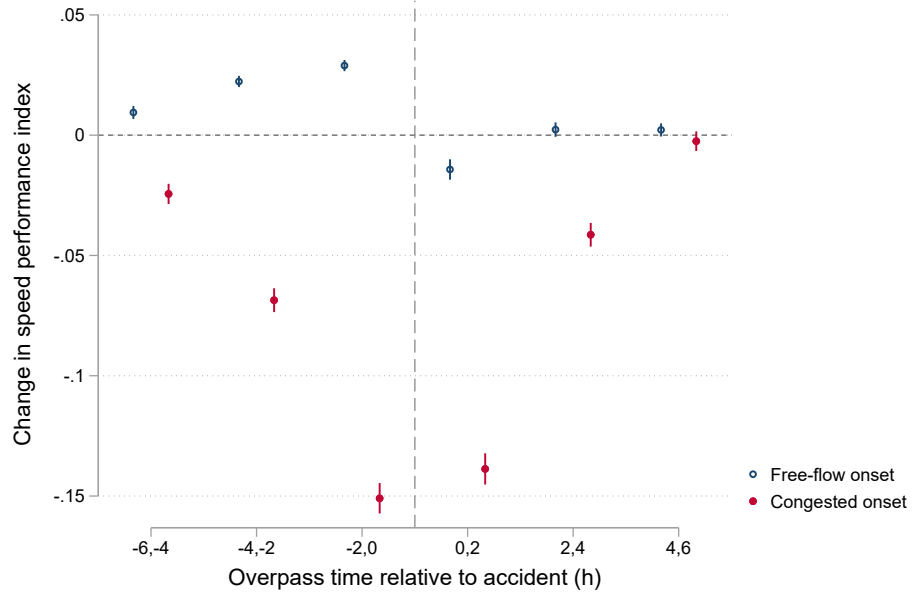


Figure 3: Event study by accident onset: speed performance index. Free-flow-onset accidents (segment free-flowing before impact) versus congested-onset accidents, estimated jointly. Congestion ramps up before congested-onset accidents (reverse causality). Before free-flow-onset accidents it does not: the series drifts slightly *upward* (more free-flowing), reflecting the persistence of the pre-state the instrument conditions on, and drops only at impact. Detector + date + hour-of-day FE; 95% CIs clustered at the detector.

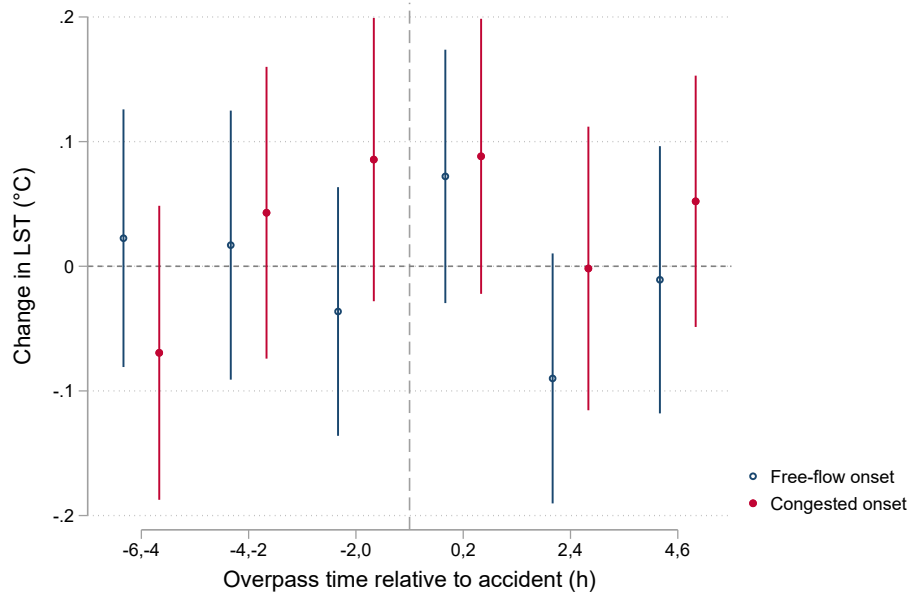


Figure 4: Event study by accident onset: surface temperature. Same specification as Figure 3, $Y = \text{LST } (^\circ\text{C})$. Pre-accident leads are flat for both groups, consistent with the exclusion restriction.

The cost is that free-flow-onset accidents perturb congestion less than the average accident, which bears on precision (Section 5).

4.3 The Free-Flow-Onset Instrument

Our instrument for SPI_{it} is an indicator for a *free-flow-onset* accident on segment i in the hours before the overpass, built in two steps. First, we flag any qualifying accident (a collision, hit-and-run collision, or spinout; Section 3.1) on segment i within a lookback window of length h , with $h = 2$ hours in the baseline. Second, we keep only accidents that struck while the segment was free-flowing, defined as a speed performance index of at least 0.90 in the hour before impact, so that congestion cannot have caused the accident:

$$Z_{it} = \mathbf{1}[\text{free-flow-onset accident on segment } i \text{ in } [t - h, t]].$$

Restricting to free-flow-onset accidents removes the reverse causality the event study revealed (Section 4.2); we report the all-accident instrument alongside so the contamination stays visible, and take the free-flow-onset estimate as preferred. The two-hour baseline matches the window over which the event study concentrates the congestion shock and its temperature response; we report $h \in \{1, 3\}$ hours as robustness (Appendix) and vary the 0.90 onset threshold in Section 6.

The instrument must also be relevant, predicting congestion on the affected segment. Accidents reduce effective capacity, slowing traffic as vehicles merge around the incident; the resulting congestion can persist for one to three hours after clearance (Cao et al., 2021; Haule et al., 2019). Free-flow-onset accidents perturb congestion less than the average accident (they strike uncongested segments), so their first stage is smaller in magnitude, though still precisely estimated; we report the first-stage F with each specification and, because it is moderate, rely on weak-instrument-robust inference for the preferred estimate rather than on the conventional t -ratio (Section 5). A severity check (Section 6) weights the instrument

by accident duration, since longer accidents sustain larger congestion. The compliers are accidents on normally uncongested segments; Section 5.4 discusses what their local average treatment effect represents.

The exclusion restriction requires that incident occurrence affects LST only through its effect on traffic congestion, conditional on the fixed effects and controls. We identify three potential violations and address each empirically; the event study in Section 4.2 provides the primary test, with the robustness checks in Section 6 providing supporting evidence.

Volume-driven clustering. Accident probability rises with traffic volume, which is correlated with congestion. The fixed effects absorb the systematic component of this relationship (Section 4.1), leaving the question of whether within-segment, within-date accident timing correlates with temperature determinants beyond what they and the weather controls capture. The event study’s flat pre-accident LST coefficients are the central piece of supporting evidence (there is no anticipatory movement in the outcome), and a prior-day placebo corroborates it (Section 6). We omit contemporaneous traffic volume because it is jointly determined with congestion and itself reduced by accidents, a bad control that would absorb part of the causal pathway (Angrist and Pischke, 2009); a volume-augmented specification (Table 4) confirms the estimate is insensitive to its inclusion.

Weather. Precipitation raises accident rates and lowers LST through evaporative cooling and reduced solar radiation. The fixed effects and weather controls $\mathbf{X}_{it}$ absorb most of this pattern, and the weather orthogonality test (Section 6) assesses the residual instrument–weather correlation.

Direct thermal effects and incident classification. Vehicle fires emit heat directly at the road surface, elevating pixel temperatures through a non-congestion channel. We exclude vehicle fires. For qualifying incidents (collisions, hit-and-run collisions, and spinouts), the crash scene, including any emergency response, is a point source: it lies within 100 meters of the matched detector for about a quarter of free-flow-onset accidents, and half of these accidents last 30 minutes or less, so an active scene sits inside the near-road disk at the time

of the overpass in only a minority of treated observations. Stopped traffic in the queue is a separate channel, which we bound in Section 5.4.

4.4 Two-Stage Least Squares Specification

The first stage regresses the speed performance index on the free-flow-onset instrument Z_{it} and the fixed effects and controls; the second stage replaces it with its fitted value:

$$\text{SPI}_{it} = \alpha_i + \mu_d + \lambda_h + \beta Z_{it} + \mathbf{X}'_{it}\gamma + \varepsilon_{it} \quad (2)$$

$$\text{LST}_{it} = \alpha_i + \mu_d + \lambda_h + \delta \widehat{\text{SPI}}_{it} + \mathbf{X}'_{it}\gamma + u_{it} \quad (3)$$

We expect $\beta < 0$: accidents raise congestion, lowering the index. The coefficient of interest, δ , is the effect of a one-unit decrease in the speed performance index (a shift from free-flow to standstill) on LST in degrees Celsius. We also report the implied effect of a 10 mph speed reduction (the unit used by Lee and Berkelhammer, 2025) and estimate the second stage for each distance band, so that δ traces the temperature response from the roadway out to the surrounding residential area (Section 5). Because a free-flow-onset accident lowers the index by only about 0.015, the full one-unit quantity is an extrapolation far beyond the identified margin (Section 5.4).

The fixed effects and weather controls are those of the population relationship in Section 4.1: identification is within-segment, within-date, and within-hour, conditional on local air temperature, precipitation, humidity, and wind.

We estimate the model with `ivreghdfe` (Baum et al., 2007) and cluster standard errors at the detector; the estimate is insensitive to two-way (detector-by-date) and to coarser spatial clustering (Section 6). Alongside conventional standard errors we report Anderson–Rubin confidence sets for the preferred estimate (Section 5), which are valid regardless of instrument strength (Andrews et al., 2019).³

³Because two-stage least squares with non-saturated covariates need not recover a positively weighted average of treatment effects (Blandhol et al., 2022), we treat δ as a local, design-based summary rather than

5 Results

The estimation sample is the full panel of roughly 3.9 million detector-overpass observations (Section 3). We present the results as a progression: the first stage, the OLS and all-accident benchmarks, the clean free-flow-onset estimate, and the distance gradient from the road to surrounding households.

5.1 First stage

Table 3 reports the first stage for three instruments: an accident in the prior two hours lowers the speed performance index by 0.077 using all accidents ($F \approx 1,293$), 0.015 using free-flow-onset accidents ($F \approx 48$), and 0.143 using congested-onset accidents ($F \approx 1,849$). Per Section 4.2, only the free-flow-onset coefficient reflects an exogenous shock, smaller because these accidents strike uncongested segments.

Table 3: First stage: freeway accidents and the speed performance index

	(1) All accidents	(2) Free-flow onset	(3) Congested onset
Accident, any onset (prior 2h)	-0.0768*** (0.00214)		
Free-flow-onset accident (prior 2h)		-0.0152*** (0.00218)	
Congested-onset accident (prior 2h)			-0.143*** (0.00333)
First-stage F (cluster-robust)	1,293.4	48.3	1,849.4
Treated (instrument=1)	10,483	5,378	5,075
Observations	3,890,495	3,885,390	3,885,087

Notes: Dependent variable is the speed performance index (speed / free-flow speed; lower = more congested). Detector (α_i), date (μ_d), and hour-of-day (λ_h) fixed effects; SE clustered at the detector. “All” uses every accident; “free-flow onset” restricts to accidents that struck while the segment was free-flowing (pre-accident SPI ≥ 0.90), which the event study shows are not preceded by a congestion pre-trend; “congested onset” restricts to accidents during existing congestion, shown to document the reverse causality. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

a structural parameter and lean on the reduced-form event-study evidence for the qualitative conclusion.

5.2 OLS and the contaminated instrument

Table 4 reports second-stage estimates. Within detector, date, and hour-of-day, the OLS coefficient on the speed performance index is $+0.14$ ($p < 0.05$): small and wrong-signed, implying that more congestion is associated with very slightly *lower* temperature. (The sign and magnitude are unchanged when the local weather controls are added, so the contrast with the IV estimates below is not an artifact of omitting weather.) We do not interpret this coefficient causally, and we do not read it as evidence against the mechanism. Congestion forms endogenously with respect to conditions that also move surface temperature, and the sign of the resulting bias is not signed *ex ante* (Section 4.1); the within-cell association is therefore not an estimate of the effect of congestion on temperature. That is what motivates the instrument. The all-accident 2SLS estimate is -1.14 ($p < 0.05$): correctly signed, precise, and contaminated (Section 4.2). Adding contemporaneous traffic volume, a potential bad control, barely moves it (-1.19); the binding problem is the instrument’s validity, not volume.

Table 4: Effect of congestion on land surface temperature: OLS and 2SLS

	(1)	(2)	(3)	(4)
	OLS	2SLS (all)	2SLS (free-flow)	2SLS (+vol)
Speed performance index	0.136**	-1.137**	-4.774	-1.187**
	(0.0590)	(0.493)	(3.477)	(0.499)
Traffic flow (1,000 veh/hr)				-0.0699***
				(0.0123)
Treated (instrument=1)	10,483	10,483	5,378	10,483
Observations	3,890,495	3,890,495	3,885,390	3,890,495

Notes: Outcome is LST (°C) in the 500-meter buffer; $\alpha_i + \mu_d + \lambda_h$ fixed effects; SE clustered at the detector. Columns: OLS; 2SLS with all accidents (contaminated by reverse causality, Section 4.2); 2SLS with free-flow-onset accidents (clean); and 2SLS adding contemporaneous volume (a potential bad control, Section 4.3). The free-flow-onset column is the binary instrument; Tables 6 and 7 refine it. Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 The clean free-flow-onset estimate

The quantity we emphasize is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.08°C over the following two hours (0.10°C without weather controls), against flat pre-accident leads (Section 4.2). The event study’s impact coefficient (+0.07°C) is the same response estimated with lead and lag bins included; the regression here pools it into a single coefficient. This is the direct experimental contrast (treated versus untreated overpasses of the same detector, date, and hour), and it requires no stand on how the instrument’s association with congestion divides between the accident shock and the selection inherent in conditioning on a free-flowing pre-hour. Table 5 reports this reduced form directly, by distance ring, without and with weather controls. We take this reduced form as our primary result.

Table 5: Reduced form: effect of a free-flow-onset accident on land surface temperature ($^{\circ}\text{C}$), by distance ring

	0–100 m (near-road)		100–500 m ring		500–1000 m ring	
	(1)	(2)	(3)	(4)	(5)	(6)
	No weather	+ Weather	No weather	+ Weather	No weather	+ Weather
Free-flow-onset accident (prior 2h)	0.0995* (0.0517)	0.0835* (0.0443)	0.0672 (0.0503)	0.0611 (0.0428)	0.0368 (0.0514)	0.0234 (0.0441)
Free-flow-onset treated	5,110	4,851	5,058	4,804	5,309	5,036
Observations	3,690,788	3,512,679	3,635,480	3,459,683	3,834,079	3,648,286

Notes: Reduced-form regressions of ring-mean LST ($^{\circ}\text{C}$) on the free-flow-onset accident indicator, on the clean sample (free-flow-onset treated and no-accident control overpasses). All columns include detector, date, and hour-of-day fixed effects; “+ Weather” columns add air temperature, relative humidity, wind speed, and precipitation at the detector and overpass hour. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The 2SLS estimates serve mainly to confirm that this response is right-signed and robust. Across instruments (binary, duration-weighted, and the two combined in an overidentified specification) and across outcome bands from 100 to 500 meters, δ is right-signed and stable in rough magnitude, tightening as the instrument carries more information and the outcome concentrates near the road (Tables 6 and 7); the weather controls $\mathbf{X}_{it}$ identify the effect over and above ambient temperature (the air-temperature coefficient alone is about $+0.77^{\circ}\text{C}$ of LST per $^{\circ}\text{C}$ of air temperature). The weather-controlled overidentified estimate at the 100-meter band is $\delta = -5.36$ (SE 2.52, $p = 0.033$, $F \approx 30$). The estimate is significant at conventional levels near the road and weakens across the wider bands ($p = 0.053$ at 250 meters, $p = 0.10$ at 500 meters, and $p = 0.17$ for the binary instrument on the 500-meter outcome, Table 6). We read this per-unit coefficient as a calibration of the reduced form (Section 5.4) rather than as the headline.

Table 6: 2SLS estimates: instrument variants and outcome refinements

	(1)	(2)	(3)	(4)
	Binary	Duration	Overidentified	Spatially demeaned
Speed performance index	-4.774	-6.021*	-5.853*	-4.048*
	(3.477)	(3.388)	(3.062)	(2.318)
KP first-stage F	48.3	32.8	28.8	10.2
Hansen J p (overid)	—	—	0.921	—
Free-flow-onset treated	5,378	5,217	5,217	5,225
Observations	3,885,390	3,885,229	3,885,229	3,811,038

Notes: 2SLS estimates of the effect of the speed performance index on LST (°C) in the 500-meter buffer, without weather controls. The instrument is the free-flow-onset accident indicator in column (1), its duration in column (2), and both (overidentified) in column (3). Column (4) uses the binary instrument with both the outcome and the speed performance index demeaned against detectors in the same three-kilometer cell observed at the same overpass, which absorbs shocks common to the neighborhood; the first stage falls because demeaning also removes part of the instrument’s variation. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We also report Anderson–Rubin confidence sets, which are valid regardless of instrument strength. For the preferred 100-meter estimate the 95% AR set is approximately $[-12.3, +0.5]$, wider than the Wald interval, with $p \approx 0.09$ against a null of zero; under this more conservative standard the near-road response is significant at the 10 percent level.

5.4 Calibrating the per-unit magnitude

To compare with the literature, we scale the reduced form into per-unit-congestion terms; this is a calibration, not the headline. The scaling depends on how much of the instrument’s association with congestion is credited to the accident shock rather than to the selection

Table 7: Weather-controlled specification grid: overidentified instruments on near-road bands

	100 m disk		250 m disk		500 m disk	
	(1)	(2)	(3)	(4)	(5)	(6)
	No weather	+ Weather	No weather	+ Weather	No weather	+ Weather
Speed performance index	-6.581** (3.015)	-5.357** (2.518)	-6.148** (3.032)	-4.974* (2.571)	-5.853* (3.062)	-4.298* (2.586)
KP first-stage F	29.5	29.7	29.3	29.3	28.8	28.9
Hansen J p (overid)	0.709	0.832	0.876	0.438	0.921	0.386
Free-flow-onset treated	4,954	4,699	5,108	4,843	5,217	4,944
Observations	3,690,632	3,512,527	3,806,820	3,623,207	3,885,229	3,697,660

Notes: 2SLS estimates of the effect of the speed performance index on LST ($^{\circ}\text{C}$) in cumulative disks of the stated radius, instrumented by the free-flow-onset accident indicator and its duration (overidentified). “+ Weather” columns add air temperature, relative humidity, wind speed, and precipitation at the detector and overpass hour. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

inherent in conditioning on a free-flowing pre-hour, and that choice must be applied to both sides of the ratio. Taking the measured quantities at face value, the 0.084°C reduced form over a first stage of 0.016 gives $\delta \approx 5.4$. Substituting the selection-adjusted first stage from the pseudo-event placebo (-0.036) requires also netting out the placebo’s own reduced form (-0.030 at 100 meters), which raises the numerator to 0.114°C and gives $\delta \approx 3.2$. At a mean free-flow speed of 68.7 mph, 10 mph is about 0.15 index units, so the two accountings imply roughly 0.5 to 0.8°C per 10 mph of slowdown. Both endpoints exceed the 0.36°C that Lee and Berkelhammer (2025) report from bus speeds in Chicago. We do not read that gap as a like-for-like comparison: their outcome is a two-kilometer heat-island average and ours a 100-meter disk dominated by the road surface, their speeds are 20–50 mph on city streets against a 68.7 mph freeway free-flow baseline, and their design is associational. We place no weight on the adjusted endpoint alone: the placebo reduced form is itself imprecise ($p = 0.53$), which is why we report the pair rather than a point. Expressed as a free-flow-to-standstill difference (about 3.2 to 5.4°C), the quantity further extrapolates a locally identified response across the full congestion range. For these reasons we read the per-unit magnitude as a calibration and anchor interpretation on the reduced form. The estimate is also a local average treatment effect for the complier population (segments that are normally free-flowing

and experience a sudden, exogenous slowdown), so it speaks to the marginal heat cost of an unusual slowdown rather than to the steady-state heat of a chronically congested freeway.

Is the near-road effect just stopped vehicles in the pixel? Because ECOSTRESS records the radiometric temperature of whatever fills a pixel, a queue of sun-heated vehicles could raise the measured near-road temperature without warming the surrounding environment. A back-of-envelope calculation (derivation in Appendix) bounds this channel: free-flow-onset accidents barely slow traffic, so they add only a few vehicles to the 100-meter buffer, and even generously hot vehicle surfaces raise the pixel mean by only about 0.01°C under the measured first stage. Vehicle composition can thus account for roughly an eighth of the near-road response under the measured accounting, and about a fifth under the selection-adjusted one, where a larger first stage implies more marginal vehicles but also a larger reduced form. On either accounting the bulk of the response is genuine warming of the road environment (0.07°C of the measured 0.084 , 0.09°C of the adjusted 0.114), consistent with the waste-heat mechanism. The same arithmetic explains why the downwind and upwind half-disks give similar estimates (Table 16): at these densities neither wind-borne dispersion nor a compositional artifact is large enough to produce a detectable asymmetry.

5.5 From the road to nearby households

Table 8 and Figure 5 report the clean estimate by distance from the freeway. This distance gradient is the paper’s spatial object—how far a localized road externality reaches into the surrounding residential space, the thermal analog of the distance decay traced for transport and environmental disamenities (Seo et al., 2014; Currie et al., 2015). We use *disjoint* annular rings—0–100 meters, 100–500 meters, and 500–1,000 meters, rather than nested disks, so that each estimate reflects the temperature response in a distinct distance zone and the gradient is not mechanically smoothed by sharing near-road pixels across bands. The near-road ring captures on-road and roadside heat; the outer ring captures the residential area around

the corridor, the exposure of most direct welfare interest. The gradient adjudicates two questions: whether a measurable effect persists out to the residential ring, and whether it decays with distance or instead rises in the intermediate zone, which would point to traffic diverted onto surface streets.

With weather controls, the gradient decays monotonically from the road outward (Table 8): $\delta = -5.36$ at the 0–100-meter ring ($p = 0.033$; identical to the preferred 100-meter estimate, since the innermost ring is that band), -4.38 at 100–500 meters ($p = 0.08$), and -2.76 at the 500–1,000-meter residential ring ($p = 0.28$). The effect is right-signed in every ring and statistically resolved near the road. The residential-ring estimate is of the expected sign and a plausible magnitude, but its confidence interval contains both zero and an effect as large as the near-road estimate; since it rests on a treated sample of similar size to the near-road ring (about 5,000 free-flow-onset overpasses), the outer ring is better described as underpowered than as null. The response is concentrated at the road and falls off with distance, with no rise in the intermediate zone of the kind a surface-street diversion effect would produce. We do not attach a physical transport mechanism to the outer rings. The estimates there are not resolved, and our design identifies where the response appears, not what carries it. Whether the decay itself is statistically real, rather than a pattern read off separate confidence intervals, is tested formally below.

Table 8: Weather-controlled distance gradient: disjoint rings (road to households)

	(1)	(2)	(3)
	0–100 m	100–500 m	500–1000 m
Speed performance index	-5.357** (2.518)	-4.383* (2.481)	-2.757 (2.566)
KP first-stage F	29.7	28.5	27.5
Hansen J p (overid)	0.832	0.688	0.309
Free-flow-onset treated	4,699	4,654	4,881
Observations	3,512,527	3,459,533	3,648,131

Notes: 2SLS estimates of the effect of the speed performance index on LST ($^{\circ}\text{C}$) in disjoint annular rings around the detector, instrumented by the free-flow-onset accident indicator and its duration (overidentified), with weather controls (air temperature, relative humidity, wind speed, precipitation). Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

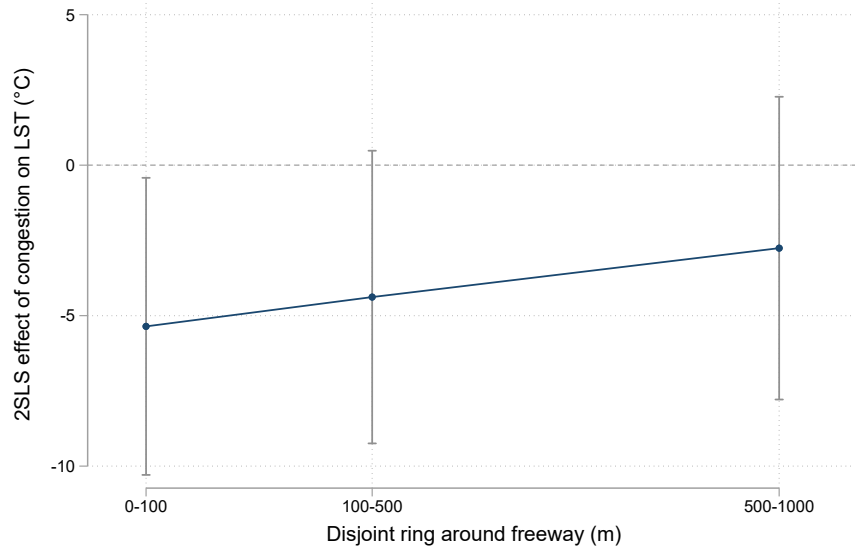


Figure 5: Congestion-heat effect by distance from the freeway. Clean free-flow-onset overidentified 2SLS estimate of δ (effect of the speed performance index on LST, $^{\circ}\text{C}$) with weather controls (the preferred specification) for disjoint annular rings 0–100, 100–500, and 500–1,000 meters from the detector, with 95% confidence intervals. Disjoint rings (rather than nested disks) ensure each estimate reflects a distinct distance zone. The outer ring covers the residential area around the corridor. Point estimates match Table 8.

Table 9 carries out that test. It estimates the three ring reduced forms jointly, stacking the disjoint rings and fully interacting the specification (fixed effects, weather controls, and

the accident indicator) with ring. Standard errors are clustered on a 5-kilometer spatial grid, so nearby detectors, and the three rings of the same detector-overpass, may be arbitrarily correlated. The test is cast in reduced form, consistent with the paper’s emphasis: the ring coefficients are the weather-controlled estimates of Table 5, in °C per accident window (positive, unlike the 2SLS δ above, which is signed per unit of the speed performance index), and the 2SLS ring estimates are their scaled counterparts. Although the per-ring confidence intervals overlap, the cross-ring *differences* are estimated much more precisely than the ring levels, because the rings of a given detector-overpass share common shocks that difference out of the comparison. The data reject equality of the three ring effects ($p = 0.016$) and reject equality of the near-road and residential rings ($p = 0.005$); a linear decline in the effect with distance from the road is likewise significant ($p = 0.006$). The distance decay is thus a statistically established feature of the reduced form, not a pattern inferred from overlapping point estimates. Because a broad area-level confounder would move all rings together rather than fade with distance from the road, the rejected equality also weighs against such confounding. This sharpens, rather than changes, the substantive reading: the externality demonstrably weakens with distance from its source, while the level of the response within the residential ring itself remains imprecisely estimated.

How many people live in these rings? Table 10 reports the residential population around the studied freeway segments from the 2018–2022 American Community Survey. About 11 million Californians live within one kilometer of these segments, and 4.7 million within 500 meters. This population is more racially diverse than the state as a whole (42 percent Hispanic and 71 percent non-white, against roughly 40 and 65 percent statewide), though its median household income is close to the state median. This characterizes the scale and composition of the exposed population; it is not a claim that the congestion-heat effect falls disproportionately on these residents, which our estimates do not support.

Table 9: Formal test of the reduced-form distance gradient (jointly estimated rings, 5 km spatial clustering)

ff-onset accident $\times$ 0–100 m (near-road)	0.0835* (0.0451)
ff-onset accident $\times$ 100–500 m	0.0611 (0.0433)
ff-onset accident $\times$ 500–1000 m (residential)	0.0234 (0.0447)
p: all three rings equal	0.016
p: near-road = residential	0.005
Distance trend ($^{\circ}\text{C}$ per 100 m)	−0.0086
p: distance trend	0.006
Observations (stacked)	10,620,648

Notes: The three disjoint rings are stacked and estimated jointly, with the fixed effects (detector, date, hour-of-day), the weather controls, and the accident indicator each fully interacted with ring; the ring coefficients therefore reproduce the weather-controlled reduced forms of Table 5. Standard errors, in parentheses, are clustered on a 5-kilometer spatial grid (945 clusters), allowing arbitrary correlation across nearby detectors and across the rings of the same detector-overpass. Equality rows report Wald tests on the joint covariance matrix; the distance-trend rows replace the ring-specific coefficients with the accident indicator interacted with ring-midpoint distance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Residential population exposed near the studied freeway segments

Distance ring	Exposed pop.	Median income	% Hispanic	% non-white
0–100 m	319,927	\$100,193	42	71
100–500 m	4,413,002	\$100,329	42	71
500–1000 m	6,273,904	\$102,833	41	70
Corridor (0–1000 m)	11,006,834	\$101,752	42	70
California	39,356,104	\$104,733	40	65

Notes: Population, median household income, and racial composition from the 2018–2022 ACS five-year block-group estimates, area-weighted into disjoint distance-to-nearest-freeway rings around the detectors in the estimation sample. Detector buffers are unioned across segments, so each resident is counted once, by distance to the nearest studied segment. “% non-white” is one minus the non-Hispanic white share; income is population-weighted across block groups. The California row is the statewide population-weighted benchmark.

6 Robustness Checks

This section reports in full the two checks most central to the instrument’s construction and validity: a prior-day placebo, and the onset-threshold sensitivity with its pseudo-event placebo. Some checks are already reported with the main results (the distance-band gradient, Section 5, and the duration-weighted and overidentified instruments, Table 6). Nine further checks are summarized briefly at the end of this section, with the supporting tables collected in Appendix. The estimate is right-signed throughout; where a check moves its magnitude, we say so and say why.

Placebo: incidents from 24 hours prior. We replace Z_{it} with an indicator for whether a qualifying incident occurred on segment i at the same time of day on the previous calendar day. If the instrument is valid, prior-day incidents should have no predictive power for today’s LST. Table 11 confirms this: a prior-day free-flow-onset accident does not predict today’s surface temperature (reduced-form coefficient -0.08°C , statistically insignificant). It carries a small, economically negligible same-sign correlation with today’s congestion (day-to-day persistence in traffic conditions) but no significant temperature effect, as a valid instrument requires.

Table 11: Robustness: prior-day placebo

	(1)	(2)
	Speed performance index	LST ($^{\circ}\text{C}$)
Prior-day ff accident	0.0085*** (0.0016)	-0.0778 (0.0516)
Prior-day treated	5,329	5,329
Observations	3,880,012	3,880,012

Notes: Regressions of the speed performance index (column 1) and 500-meter LST (column 2) on an indicator for a free-flow-onset accident at the same time of day on the previous calendar day, estimated on the no-accident sample. A valid instrument requires no temperature effect of the prior-day accident conditional on today’s conditions. Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Free-flow-onset threshold sensitivity and pseudo-event placebo. The instrument con-

ditions on a lagged value of the endogenous variable: an accident qualifies only if its segment was free-flowing ($\text{SPI} \geq 0.90$) in the hour before impact. Because this selects on transitory traffic states, the choice of threshold matters in two directions. Loosening it readmits accidents that congestion itself may have caused, reintroducing the reverse causality the restriction exists to remove; tightening it conditions on increasingly unusual hours, raising the worry that the selected moments are transitorily atypical in temperature as well, so that mean reversion in the outcome masquerades as a treatment response. Table 12 re-estimates the preferred specification at onset thresholds of 0.85, 0.90 (the baseline), and 0.95. The estimate is right-signed at every threshold, but its magnitude is sensitive to the choice: -2.1 and insignificant at 0.85, -5.4 at the baseline, and -11.7 ($p = 0.06$) at 0.95, with the first-stage F falling from 72 to 30 to 7 as the treated sample shrinks toward the most strictly free-flowing segments. The strict end is weakly identified ($F = 7.3$), so its point estimate should not be read at face value. Against the mean-reversion reading, the event-study temperature leads for free-flow-onset accidents are flat individually and jointly ($p = 0.86$; Section 4.2): treated overpasses are not selected on transitorily warm or cool moments, so there is no temperature reversion for the post-accident bin to inherit.

We test the selection channel directly with a placebo that replicates the instrument’s construction without an accident. For each real free-flow-onset accident we assign a pseudo-event at the same timestamp to a randomly drawn other detector that was free-flowing in the pre-hour ($\text{SPI} \geq 0.90$) and had no accident within six hours; the 283,236 pseudo-events reproduce the instrument’s date and time-of-day distribution almost exactly. Table 13 reports their effects on the no-accident sample. The pseudo first stage is *positive* ($+0.021$, $t = 20$): a segment selected on a free-flowing pre-hour remains more free-flowing than its baseline over the following two hours—it persists; it does not revert below. The accident-induced drop to 0.015 *below* baseline therefore cannot be mean reversion, and measured against this no-accident counterfactual the accident’s causal effect on congestion is about -0.036 rather than the -0.015 level difference. The pseudo reduced forms on temperature

are null at both bands (-0.030 at 100 meters, $p = 0.53$; $+0.008$ at 500 meters, $p = 0.86$): the selection inherent in the instrument does not move surface temperature, which is direct support for the exclusion restriction. The placebo thus resolves the artifact question (the temperature response is caused by accidents, not by the conditioning) while making precise the first-stage accounting choice that motivates reporting the per-unit magnitude as a range (Section 5.4).

What the threshold gradient does establish is that the magnitude is not threshold-invariant. The sign and the qualitative conclusion are robust to the onset definition, but the point estimate is not, which reinforces our reading of the magnitude as informative at the identified margin rather than sharply estimated (Section 5.4).

Table 12: Robustness: free-flow-onset threshold sensitivity

	(1) SPI ≥ 0.85	(2) SPI ≥ 0.90	(3) SPI ≥ 0.95
Speed performance index	-2.089 (1.654)	-5.357** (2.518)	-11.748* (6.267)
KP first-stage F	72.5	29.7	7.3
Free-flow-onset treated	5,853	4,699	2,990
Observations	3,513,681	3,512,527	3,510,818

Notes: The preferred specification (overidentified free-flow-onset instruments, 100-meter LST outcome, weather controls) re-estimated with the onset threshold, the pre-accident speed performance index above which an accident qualifies as free-flow-onset, set to each stated value (0.90 is the baseline). Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Spatial clustering of standard errors. The baseline clusters standard errors at the detector. Because adjacent detectors can share ECOSTRESS pixels, we re-cluster at progressively coarser spatial units (roughly 5- and 10-kilometer grid cells and the freeway-by-direction corridor) that let nearby detectors correlate arbitrarily within a satellite scene (Table 14). The standard error rises only modestly, from 2.52 at the detector level to 2.73

Table 13: Placebo: pseudo free-flow-onset events (selection without an accident)

	(1) Speed performance index	(2) LST 100 m (°C)	(3) LST 500 m (°C)
Pseudo free-flow-onset event	0.0208*** (0.0010)	-0.0297 (0.0476)	0.0082 (0.0476)
Pseudo-event treated	5,693	5,406	5,693
Observations	3,692,716	3,507,828	3,692,716

Notes: Each real free-flow-onset accident is assigned a pseudo-event at the same timestamp on a randomly drawn other detector that was free-flowing in the pre-hour and had no accident within six hours, replicating the instrument’s construction without an accident. Columns report regressions of the stated outcome on the pseudo-event indicator, estimated on the no-accident sample. Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

at the corridor level; the estimate remains significant at five percent under grid-cell clustering and sits at the margin ($p = 0.05$) under the most conservative corridor clustering, so cross-detector spatial correlation does not materially understate the uncertainty. (Two-way detector-by-date clustering leaves the standard error unchanged at 2.52.)

Table 14: Spatial standard errors for the preferred estimate

	(1) Detector	(2) 5 km grid	(3) 10 km grid	(4) Corridor
Speed performance index	-5.357** (2.518)	-5.357** (2.494)	-5.357** (2.579)	-5.357* (2.733)
Clusters	8,852	945	464	194
Observations	3,512,527	3,512,527	3,512,527	3,512,527

Notes: The preferred estimate (overidentified free-flow-onset instruments, 100-meter LST outcome, weather controls, detector, date, and hour-of-day fixed effects) with standard errors clustered at progressively coarser spatial units: the detector (baseline), roughly 5- and 10-kilometer grid cells, and the freeway-by-direction corridor. The point estimate is identical by construction; only the standard error changes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Additional checks. Nine further checks, with supporting tables in Appendix, leave the conclusions intact. (1) Regressing the instrument on the weather controls shows that warmer and more humid hours see slightly more free-flow-onset accidents (Table 15), which is why the main specification controls for weather. Doing so lowers the near-road reduced form from 0.10 to 0.08°C, and the flat pre-accident temperature leads (Section 4.2) show no tem-

perature anomaly in the hours before accidents. (2) The duration-weighted instrument and the overidentified combination in Table 6 probe accident severity; the Hansen J test does not reject ($p = 0.92$), an internal-consistency check we read cautiously given its low power. (3) Splitting the 500-meter buffer into downwind and upwind half-disks by ERA5-Land wind direction, following designs that identify near-road exposure from wind direction relative to the source (Anderson, 2020; Herrnstadt et al., 2021), yields no significant differential (-3.97 vs. -5.13), so we do not lean on a thermal-dispersion interpretation (Table 16). (4) Look-back windows of one and three hours, against the two-hour baseline, leave the estimate right-signed and comparable (-3.5 and -5.0), with the first-stage F falling as the window lengthens (Table 17). (5) Dropping observations on segments within two detector stations of an accident, to remove diverted-traffic contamination of the controls, leaves the estimate unchanged (-5.88 ; Table 18). (6) Spatial demeaning removes any shock common to detectors in the same three-kilometer cell at a given overpass. The demeaning neighborhood is not a set of untreated controls: 87 percent of a detector’s cell-mates lie on the same freeway, at a median distance of 938 meters, so they are themselves partly treated, and the congestion measure must be differenced over the same neighborhood as the outcome. Demeaning both gives -4.05 (SE 2.32), close to the undemeaned estimate, at the cost of a first stage that falls to $F \approx 10$ because demeaning removes much of the instrument’s variation (Table 6, final column); a leave-one-out construction, excluding each detector from its own cell mean, gives -3.95 . (7) Tightening the detector data-completeness threshold from 50 to 90 percent leaves the near-road estimate essentially unchanged (-5.7 ; Table 19). (8) Alternative LST aggregations, the nearest single pixel (-5.0) and the within-buffer maximum (-5.49 , $p = 0.05$), match the mean-based headline (Tables 20 and 21). (9) Adding the jointly determined traffic-volume control, a potential bad control, moves the congestion estimate negligibly (Table 4).

7 Conclusion

We estimate the short-run effect of freeway congestion on nearby surface temperature in California, using freeway accidents as an instrument for congestion and ECOSTRESS land surface temperature, matched to congestion at the same hour, as the outcome. The main lesson concerns identification, and it travels beyond this application. Accidents are treated as exogenous instruments, yet they are not exogenous to congestion: an event study shows that congestion rises in the hours before a typical accident, so an instrument built from all accidents embeds reverse causality and yields a precise but contaminated estimate. Wherever events are used as instruments for the very conditions that make those events more likely, the same bias arises; conditioning on the pre-event state (here, restricting to accidents that strike a free-flowing road) is a transparent remedy, at the cost of a smaller first stage.

Under this clean design the estimated effect is correctly signed at every distance band. The central estimate is the reduced form: a free-flow-onset accident raises near-road surface temperature by about 0.08°C over the following two hours, against flat pre-trends. Expressed per unit of congestion, the response implies roughly 0.5 to 0.8°C per 10 mph of speed reduction, somewhat above the associational estimate of Lee and Berkelhammer (2025). We anchor the magnitude on this locally identified margin and report the full specification grid rather than a single number. The effect decays with distance from the freeway (a decline that is statistically significant when the rings are estimated jointly), and it remains right-signed in the outer ring that contains nearby residential areas.

The estimates bear on a concrete policy margin. Because they are identified from accident-induced episodes, they measure what incident detection and clearance programs mitigate: beyond delay and emissions, each episode imposes a measurable heat dose on the road and its immediate surroundings—a thermal cost absent from standard accounts of congestion externalities.

The estimates have limitations. The per-unit magnitude is identified from a modest congestion shock, so it is not tightly pinned down, and the temperature response in the

residential ring is not statistically resolved. Our evidence speaks to where the response is, not to what carries it: the gradient shows no rise of the kind surface-street diversion would produce, but we do not establish a physical channel by which roadway heat would reach the outer rings, and we do not claim one. The estimates are local to California freeway segments with PeMS detectors and to the variation induced by free-flow-onset accidents, and need not extend to other settings or to longer-run or larger congestion changes.

Within those bounds, the paper contributes a credibly identified estimate of a relationship that prior work could only document associationally, and a transparent way to handle reverse causality in accident-based instruments. It also contributes a spatial measure: resolving the response in distance bands locates how far a roadway heat externality reaches relative to where people live (some 11 million Californians live within a kilometer of the studied segments), which speaks to residential proximity to freeways and the spatial distribution of urban heat exposure. Whether the near-road and household temperature responses documented here are large enough to matter for health or welfare, and how they compare to the cost of congestion relief, are questions we leave to future work.

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Appendix Additional Figures and Tables

Figures 6 and 7 show the all-accident event study discussed in Section 4.2: congestion rises in the hours before a typical accident (reverse causality), while surface-temperature leads are flat. The remaining inputs are the supporting tables for the additional checks summarized in Section 6. (The spatial-demeaning and traffic-volume checks are reported with main-text tables, Table 6 and Table 4, and so have no separate table here.)

Vehicle-composition bound: derivation. Section 5.4 bounds the contribution of stopped vehicles to the measured near-road response. Under the measured first stage, a free-flow-onset accident lowers the speed performance index by about 0.015, which by the standard speed-density relation ($k/k_{\text{jam}} = 1 - \text{SPI}$) corresponds to a density increase of about $0.015 k_{\text{jam}}$. At a jam density of roughly 140 vehicles per kilometer per lane over five lanes, that is about ten extra vehicles per kilometer, or two within the ~ 200 -meter freeway segment that crosses the 100-meter buffer. At a projected area of about 10 m^2 each, these add $\sim 20 \text{ m}^2$ of vehicle surface to the buffer's $\sim 31,000 \text{ m}^2$ (70-meter pixels), a coverage increase of under 0.1 percent. Even if those surfaces are a generous 20°C hotter than their surroundings, the implied rise in the pixel-mean temperature is about 0.01°C , roughly an eighth of the measured 0.084°C reduced form. Applying the selection-adjusted accounting consistently, the larger first stage (about -0.036 , from the pseudo-event placebo) yields about five extra vehicles and an artifact near 0.024°C , but the matching reduced form also rises to about 0.114°C once the placebo's own reduced form is netted out, so the bound is near a fifth of the response rather than an eighth.

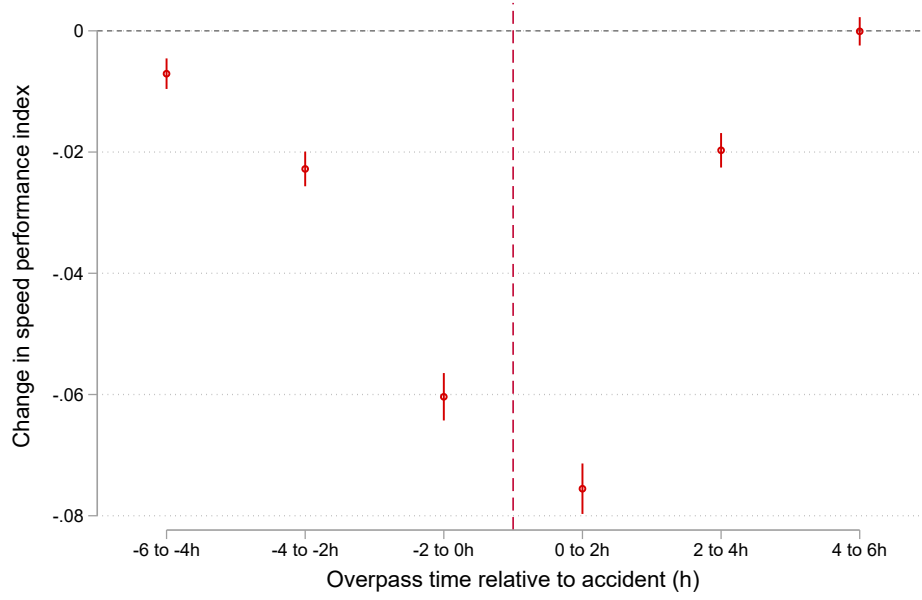


Figure 6: Event study, all accidents: speed performance index by time relative to the accident. Estimated $\hat{\phi}_{\ell,u}$ from equation (1), $Y = \text{SPI}$. Detector + date + hour-of-day fixed effects; 95% CIs clustered at the detector; baseline = no accident within ± 6 hours. Congestion rises before the accident (negative leads), indicating reverse causality.

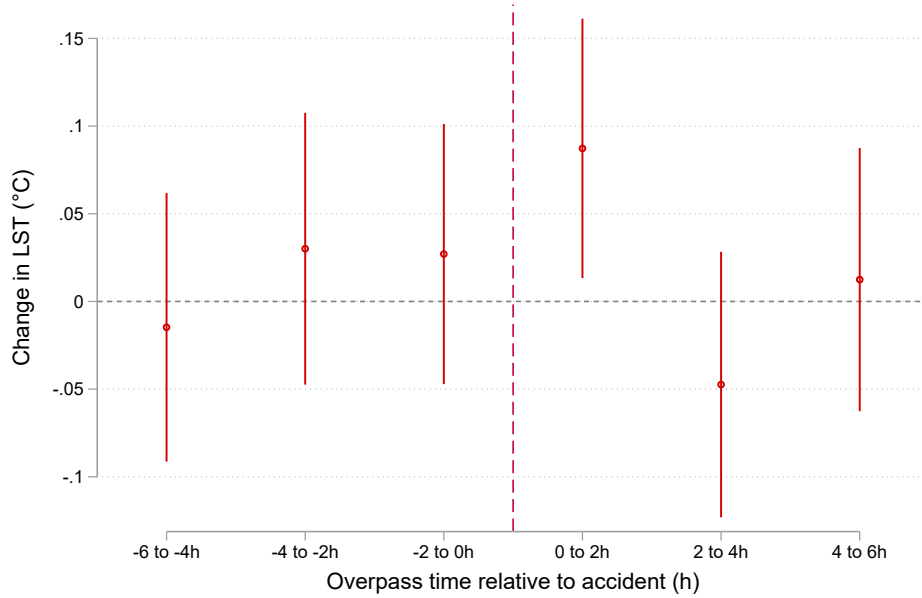


Figure 7: Event study, all accidents: surface temperature by time relative to the accident. Same specification as Figure 6, $Y = \text{LST (°C)}$. Pre-accident leads are flat and insignificant; the only significant response is at the accident bin.

Table 15: Weather orthogonality of the free-flow-onset instrument

	(1) Free-flow-onset accident
Air temperature (°C)	2.58** (1.21)
Relative humidity (%)	0.63** (0.28)
Wind speed (m/s)	-4.83 (2.97)
Precipitation (mm)	1.80 (2.85)
Observations	3,697,817
Within R-squared	0.00002

Notes: Regression of the free-flow-onset accident indicator on the weather controls, on the clean sample (free-flow-onset treated and no-accident controls). Coefficients and standard errors are multiplied by 10^5 for readability; the instrument's base rate is about 1.3×10^{-3} ; the four controls are jointly significant ($F = 2.73$, $p = 0.03$). Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 16: Robustness: downwind versus upwind 500-meter half-disks

	(1) Downwind	(2) Upwind
Speed performance index	-3.974 (2.563)	-5.129** (2.606)
KP first-stage F	29.1	28.9
Free-flow-onset treated	4,908	4,906
Observations	3,667,658	3,667,191

Notes: 2SLS estimates (overidentified free-flow-onset instruments, weather controls) on mean LST (°C) in the downwind and upwind halves of the 500-meter buffer, split by ERA5-Land wind direction at the overpass hour. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 17: Robustness: lookback window sensitivity

	(1) $h = 1$ hour	(2) $h = 3$ hours
Speed performance index	-3.545 (3.494)	-5.011 (4.088)
KP first-stage F	44.2	36.0
Free-flow-onset treated	2,594	7,658
Observations	3,688,272	3,690,801

Notes: 2SLS estimates of the effect of the speed performance index on 100-meter LST ($^{\circ}\text{C}$), with the binary free-flow-onset instrument defined over a one- and a three-hour lookback window before the overpass (baseline: two hours); no weather controls. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 18: Robustness: adjacent-segment exclusion

	(1) Full sample	(2) Drop ± 2 neighbors
Speed performance index	-5.357** (2.518)	-5.876** (2.658)
KP first-stage F	29.7	26.7
Free-flow-onset treated	4,699	4,558
Observations	3,512,527	3,500,300

Notes: The preferred specification (overidentified free-flow-onset instruments, 100-meter LST outcome, weather controls), estimated on the full sample and dropping control observations within two detector stations of an accident, which removes potential diverted-traffic contamination of the controls. Detector, date, and hour-of-day fixed effects; standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 19: Robustness: detector data completeness tightened to 90 percent

	(1) 100 m	(2) 500 m
Speed performance index	-5.661* (3.122)	-4.541 (3.249)
KP first-stage F	56.8	55.0
Free-flow-onset treated	4,257	4,489
Observations	3,136,191	3,302,563

Notes: 2SLS estimates of the effect of the speed performance index on LST (°C) at the stated band, with the binary free-flow-onset instrument and no weather controls, restricting to detector-hours with at least 90 percent of five-minute intervals observed (baseline: 50 percent). Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20: Robustness: nearest-pixel LST aggregation

	(1) Nearest pixel (°C)
Speed performance index	-4.970 (3.535)
KP first-stage F	43.9
Free-flow-onset treated	4,828
Observations	3,470,881

Notes: 2SLS estimate with the binary free-flow-onset instrument and no weather controls, replacing the buffer-mean outcome with the LST of the single ECOSTRESS pixel nearest the detector. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 21: Robustness: maximum LST in the 500-meter buffer

	(1) Maximum LST (°C)
Speed performance index	-5.487** (2.796)
KP first-stage F	28.9
Hansen J p (overid)	0.465
Free-flow-onset treated	4,944
Observations	3,697,641

Notes: 2SLS estimate (overidentified free-flow-onset instruments, weather controls) replacing the buffer-mean outcome with the maximum quality-passing pixel LST in the 500-meter buffer. Detector, date, and hour-of-day fixed effects; the sample pairs free-flow-onset-treated overpasses with no-accident controls. Standard errors, clustered by detector, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.